

The Agent Identity Framework (AIF)

Composable Accountability for AI Agents in the Digital Economy

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The gap. AI agents are beginning to place orders, sign contracts, commit code, vote in governance, and move money on behalf of people and organizations. The internet’s identity and accountability primitives — OAuth, card chargebacks, KYC, court-adjudicated tort — were designed for human principals. They do not extend cleanly to software that acts, delegates, copies itself, and upgrades continuously. Authorization is all-or-nothing. Accountability is ambiguous. Reversibility is implicit and inconsistent across rails.

What AIF is. A composable accountability layer between the identity plane (W3C DIDs, Verifiable Credentials, UCAN, OpenID SSF/CAEP) and the settlement plane (x402, MPP, ACP, UCP, card networks, stablecoin escrow). AIF does not replace either. It contributes the accountability surface between them — the part today’s landscape leaves underdefined.

Three minimum artifacts.

1. **Action Envelope** — a canonical signed wrapper for every binding action, carrying the authority formula, a structured reversibility descriptor, the settlement reference, and the accountability trustee.
2. **Accountability Escrow** — a bonded trustee role that holds a sealed pointer to the liable root principal. Counterparties see who would be liable, not necessarily who the end user is, in the happy path. A valid dispute trigger unseals the pointer.
3. **Receipt protocol** — append-only transparency-log receipts with Merkle inclusion proofs, optionally anchored on-chain for high-value flows. Selective redaction honors deletion obligations without invalidating commitments.

Identity binds to promises, not weights. An agent’s DID commits to a *behavior contract* and a *harness attestation* (the TEE-rooted policy engine that enforces the contract), plus an optional model-provenance credential from the vendor. Fine-tuning and continuous training do not churn identities; counterparties get a crisp statement of what the agent promised to do.

Authority is an algebra. Capability tokens compose with AND, OR, THRESHOLD(k of n), and WITHIN(caveat) — covering joint principals, multi-tenant operators, and cross-org hops natively. Interoperable with UCAN.

Reversibility is a typed descriptor, not a letter grade. Fields: window, recoverable value, cooperation requirement, external visibility, compensation policy reference. Shipped-and-returnable and published-and-retractable differ on the right axes; a coffee refund and a million-dollar return no longer share a class.

Liability is honest. A protocol cannot bind a court. AIF specifies a **framework-membership allocation** between consenting participants — operator first-loss → reputation bond → per-task bond → insurance pool → residual to root principal — that does not preempt tort law, regulatory rules, or consumer-credit carve-outs.

Five conformance levels. - **L0** — Envelope + authority algebra. Minimum interoperability. - **L1** — + Accountability Escrow. Enables pseudonymous consumer flows. - **L2** — + Enforced reversibility + compensation policy. Cross-rail rollback. - **L3** — + Bonded issuers + anchored receipts. High-value B2B. - **L4** — + Reputation + insurance pool. Consumer commerce at scale.

Nested: every L_n is an $L_{(n-1)}$. Adopters start small and advance under commercial pressure.

Composes with everything shipping now. Explicit integration seams for x402, MPP, ACP/UCP, A2A, MCP, Microsoft Agent Mesh / IATP, ZeroID, UCAN, SSF/CAEP, Sigstore/Rekor, EAS, and TEE attestations (SEV-SNP, TDX, Nitro).

Roadmap. White paper v1.0 published April 2026. Target: Envelope v0 reference implementation (Go, TypeScript) + Accountability Escrow stub + one integration adapter within six months. Two pilot domains: corporate procurement and regulated fiduciary flows. Open governance under Linux Foundation AIF.

Open calls.

- **Design partners** — enterprises in corporate procurement or regulated fiduciary flows willing to pilot L1 or L2 in Q3 2026. LOI, not contract.

- **Co-builders** — one DID/PKI engineer, one payments/escrow engineer, for a twelve-week reference implementation sprint.
- **Standards engagement** — NIST CAISI AI Agent Standards Initiative, Linux Foundation AAIF, W3C CCG, IETF SPICE, UCAN WG.
- **Advisors** — DID community, enterprise payments, insurance underwriting, agent-safety governance.

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Full paper: *The Agent Identity Framework — Composable Accountability for AI Agents in the Digital Economy* (v1.0, April 2026).